

Executive Summary

2022-01-01 → 2026-05-31 · 38,688 1H bars · Scenario: Session 08-21 · OI signal:

✓ Real basis (premiumIndex)

+271.3%

Total Return

9.720

Sharpe Ratio

-74.0%

Max Drawdown

3.67

Calmar Ratio

55.5%

Win Rate

1.15

Profit Factor

1,855

Trades

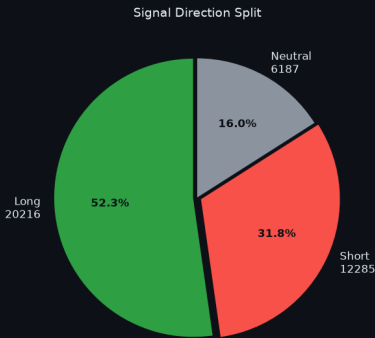
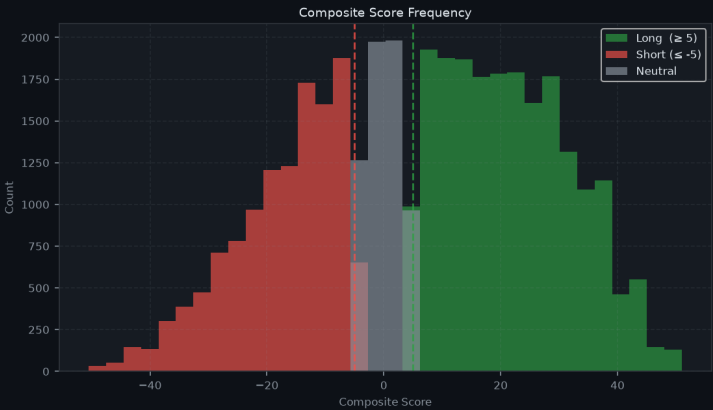
\$371,334

Final Equity

Signal Analysis

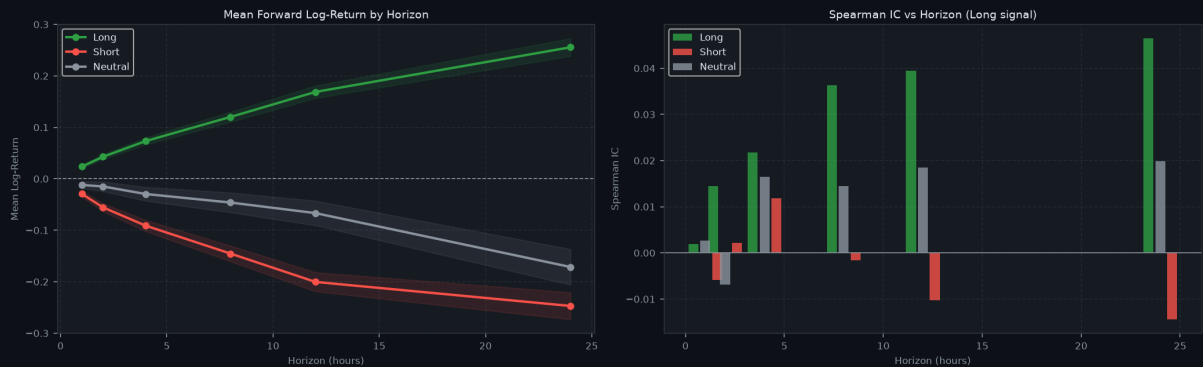
SCORE DISTRIBUTION

A1 · Signal Score Distribution



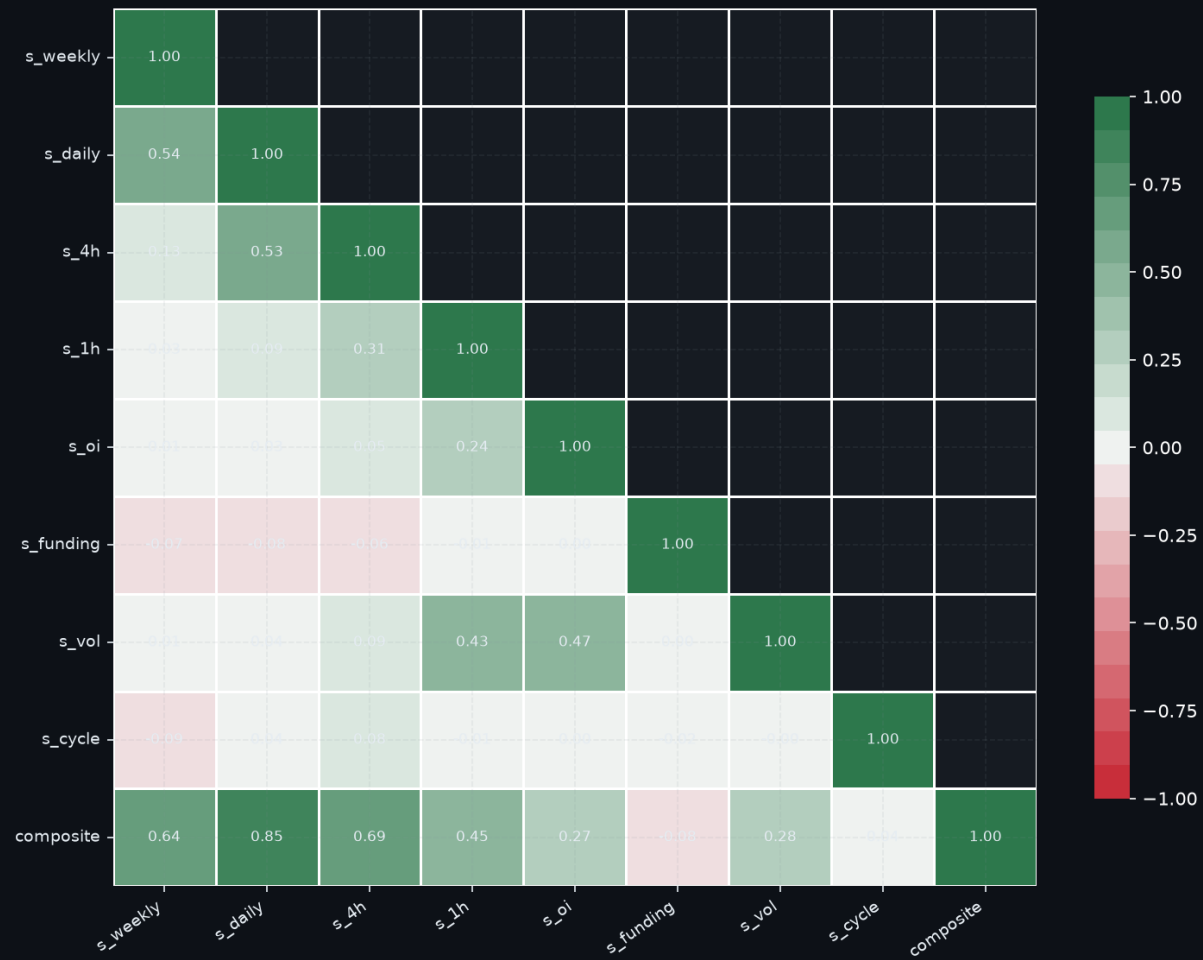
ALPHA DECAY

A2 · Alpha-Decay Analysis



COMPONENT CORRELATION

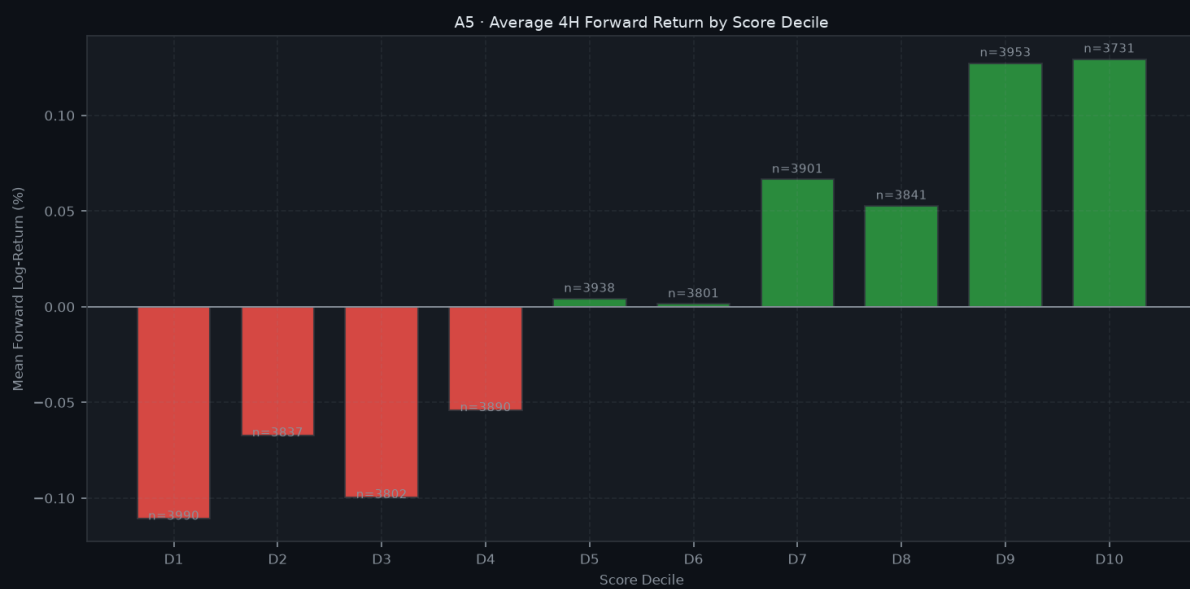
A3 · Signal Component Correlation Matrix



SCORE VS FWD RETURN



QUANTILE RETURNS



MARKET OVERVIEW

B1 · BTCUSDT Daily Market Overview (Price + EMA Stack + Volume)



OI / BASIS ANALYSIS

B2 · Open Interest Analysis + Funding Rate



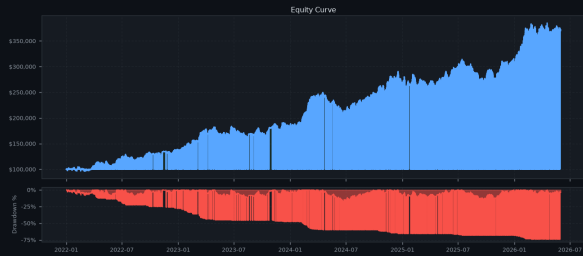
CYCLICALITY

B3-4 · BTC Seasonality & Cyclical Patterns



Backtest Performance — Full Dataset (In-Sample)

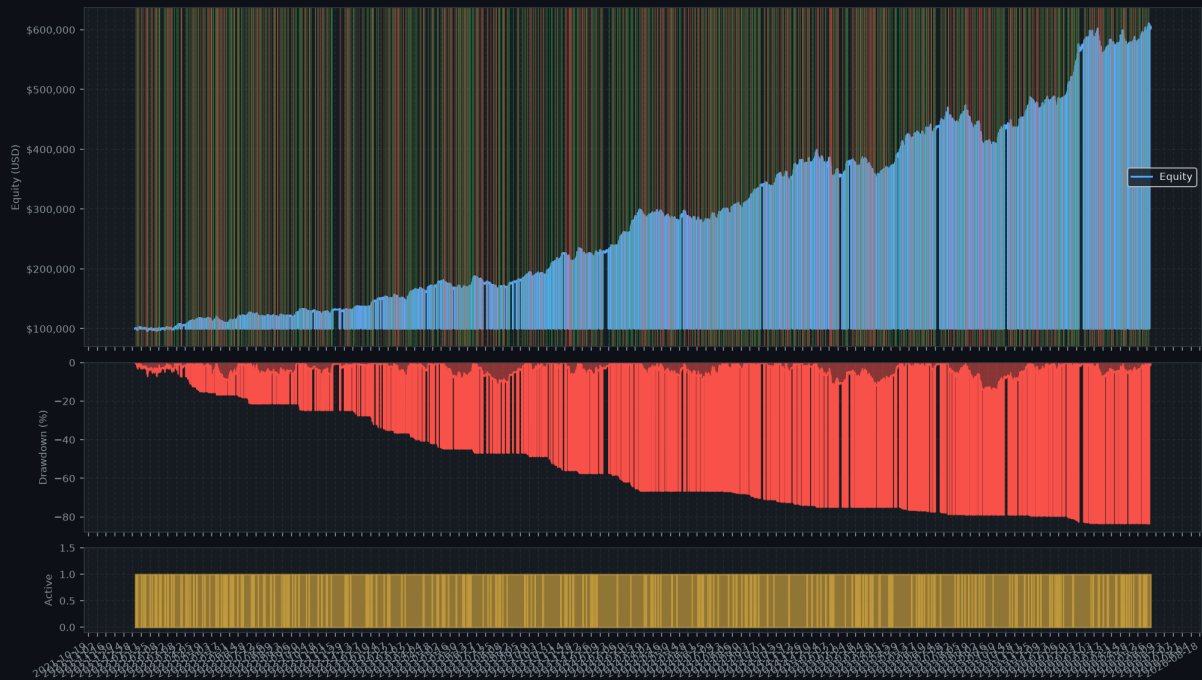
Metric	Value
Total Return	+271.33%
Sharpe Ratio	9.720
Sortino Ratio	19.589
Calmar Ratio	3.665
Max Drawdown	-74.03%
Win Rate	55.5%
Profit Factor	1.15
# Trades	1,855
Avg Win	\$+2,065.33



Metric	Value
Avg Loss	\$-2,249.78
Expectancy	+\$146.21
Avg Duration	17h

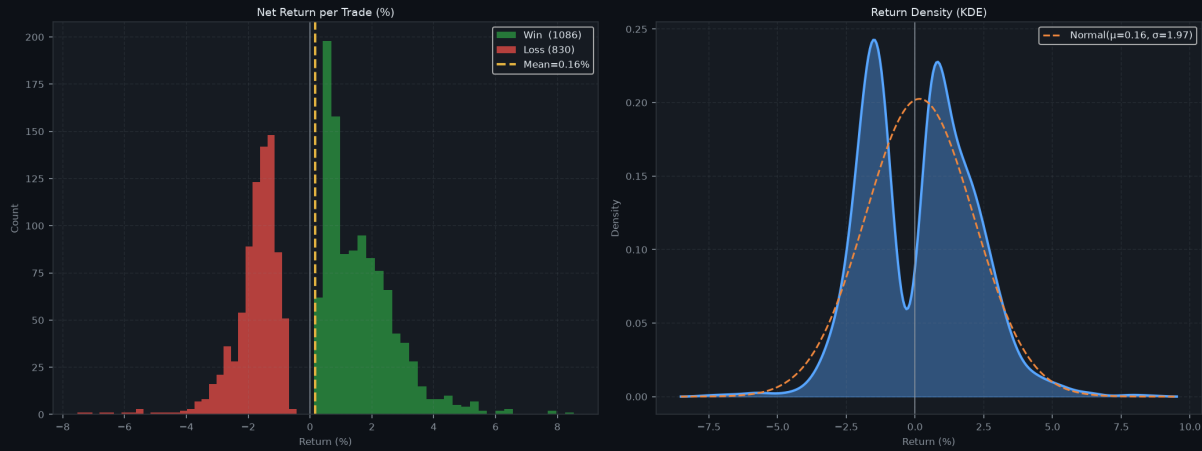
EQUITY & DRAWDOWN DETAIL

C1 · Strategy Equity Curve (BTCUSD Multi-TF)

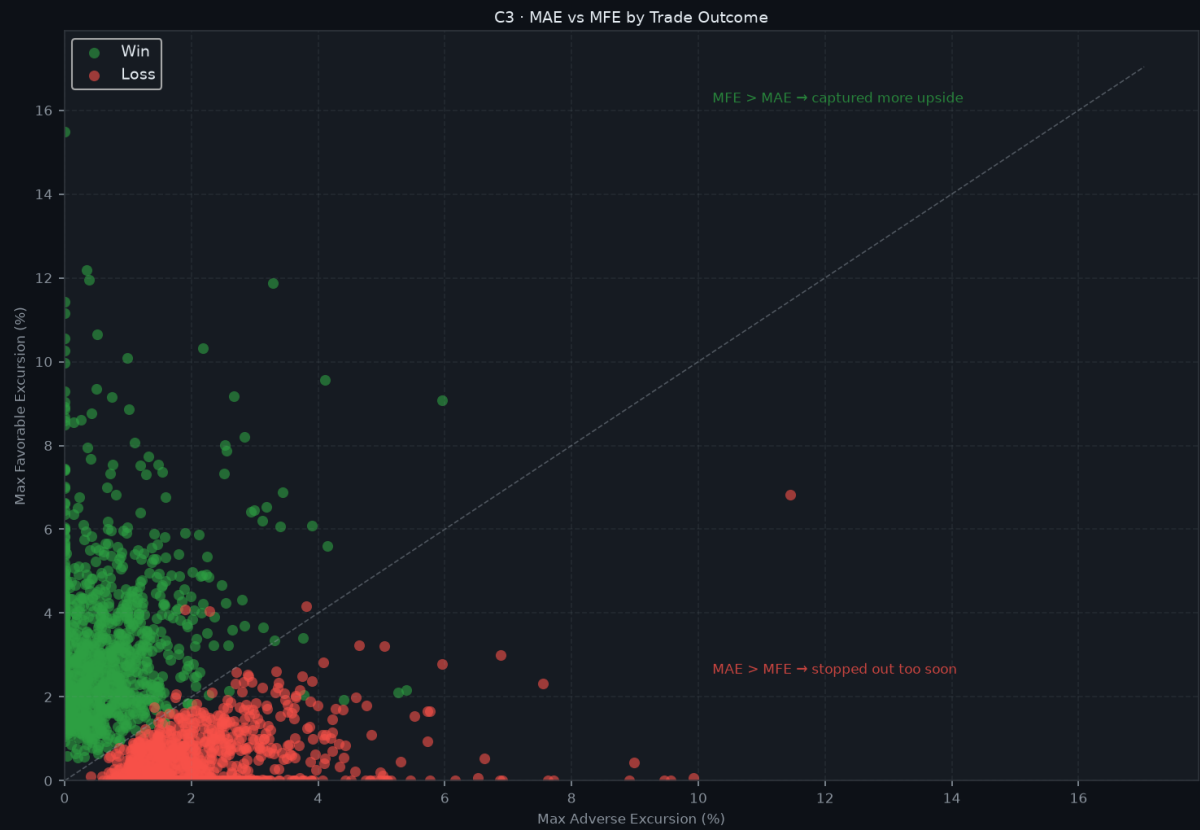


TRADE DISTRIBUTION

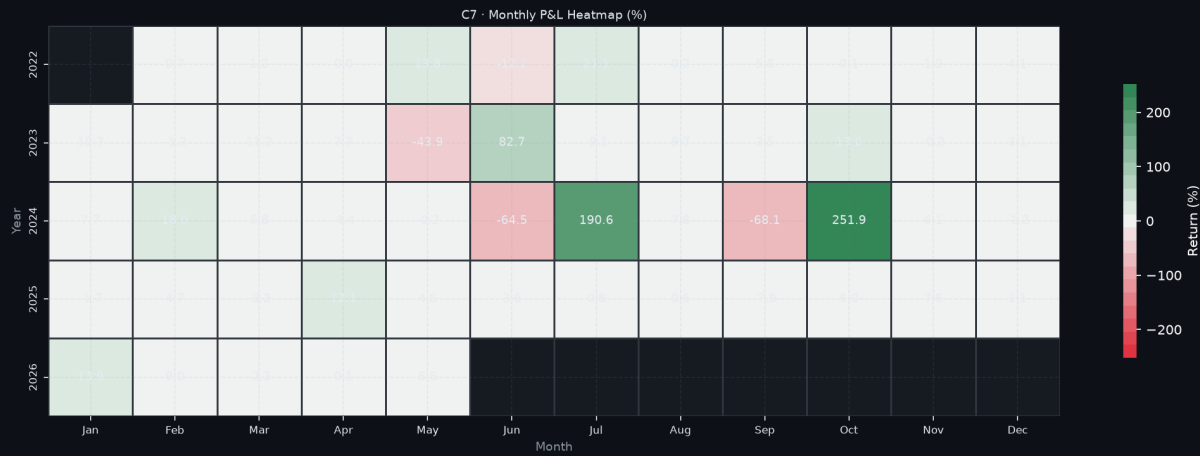
C2 · Trade Return Distribution



MAE VS MFE

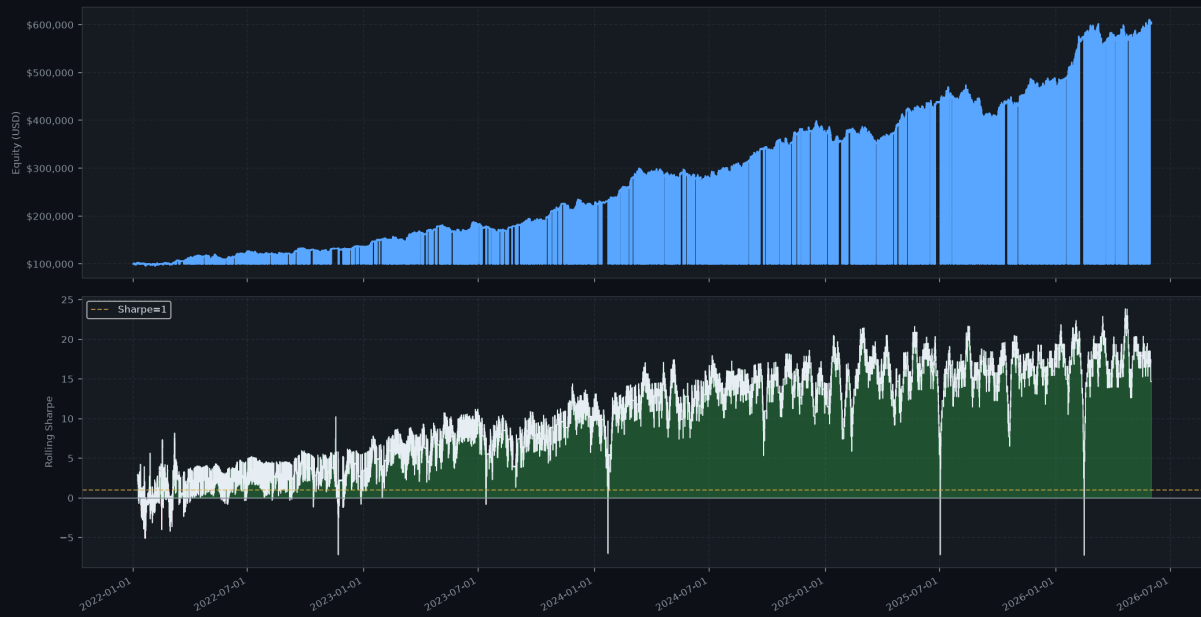


MONTHLY PNL HEATMAP



ROLLING SHARPE

C6 · Rolling 7-Day Sharpe Ratio



LAST 20 TRADES

Entry	Exit	Dir	Entry \$	Exit \$	Net PnL	Duration
2026-05-15 09:00	2026-05-15 13:00	-1	\$80,606	\$78,718	\$ +5,255.47	4h
2026-05-15 14:00	2026-05-18 06:00	-1	\$78,728	\$76,658	\$ +5,358.39	64h
2026-05-18 09:00	2026-05-18 12:00	-1	\$77,001	\$77,717	\$-3,624.36	3h
2026-05-18 13:00	2026-05-18 15:00	-1	\$77,600	\$76,084	\$ +4,921.16	2h
2026-05-18 16:00	2026-05-19 00:00	-1	\$76,394	\$77,278	\$-4,051.58	8h
2026-05-19 09:00	2026-05-19 19:00	-1	\$77,016	\$77,016	\$ +1,540.89	10h
2026-05-19 20:00	2026-05-20 08:00	-1	\$76,780	\$77,583	\$-4,025.03	12h
2026-05-20 12:00	2026-05-21 00:00	-1	\$77,351	\$77,984	\$-3,179.67	12h
2026-05-21 09:00	2026-05-21 17:00	-1	\$77,888	\$77,888	\$ +1,316.86	8h
		-1	\$77,354	\$75,581		27h

Entry	Exit	Dir	Entry \$	Exit \$	Net PnL	Duration
2026-05-21 19:00	2026-05-22 22:00				\$ +5,300.20	
2026-05-23 09:00	2026-05-23 14:00	-1	\$74,655	\$75,386	\$-3,784.02	5h
2026-05-23 15:00	2026-05-23 20:00	-1	\$75,500	\$76,187	\$-3,498.42	5h
2026-05-24 11:00	2026-05-24 14:00	1	\$77,163	\$76,446	\$-3,534.23	3h
2026-05-24 15:00	2026-05-24 23:00	-1	\$76,428	\$77,170	\$-3,642.29	8h
2026-05-25 09:00	2026-05-26 10:00	-1	\$77,422	\$77,422	\$ +1,315.17	25h
2026-05-26 11:00	2026-05-26 14:00	-1	\$77,399	\$77,399	\$ +1,282.01	3h
2026-05-26 15:00	2026-05-27 13:00	-1	\$76,874	\$75,060	\$ +5,213.55	22h
2026-05-27 14:00	2026-05-28 03:00	-1	\$75,054	\$73,510	\$ +5,133.69	13h
2026-05-28 09:00	2026-05-29 17:00	-1	\$73,294	\$74,197	\$-3,986.59	32h
2026-05-29 18:00	2026-05-31 23:00	-1	\$73,955	\$73,653	\$+978.57	53h

Walk-Forward Validation (6m train · 2m OOS · 2m step)

23

Windows

70%

Profitable

+3.95%

Median OOS Ret

2.773

OOS Sharpe

-36.8%

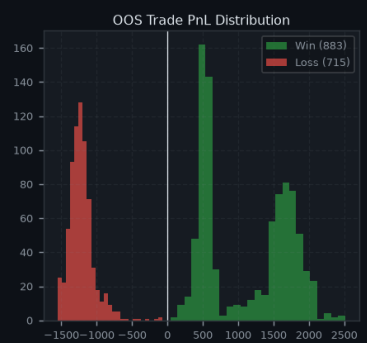
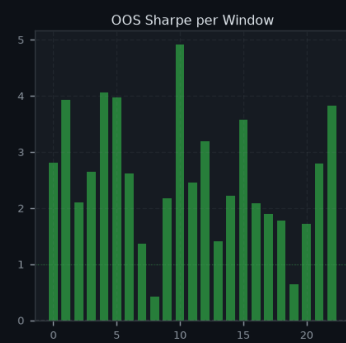
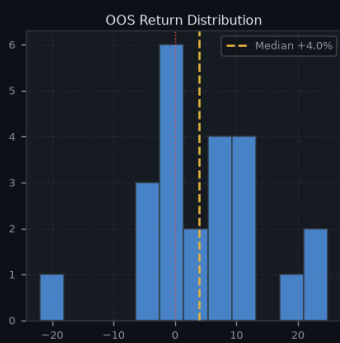
OOS Max DD

+151.6%

OOS Return

+0.464

Consistency



PER-WINDOW KPIS

Win	Train End	OOS End	OOS Ret	Trades	WR	PF	Sharpe	Max DD
W1	2022-07-01	2022-09-01	-1.6%	75	54.7%	0.95	+2.812	-22.55%
W2	2022-09-01	2022-11-01	+6.9%	70	60.0%	1.22	+3.923	-27.18%
W3	2022-11-01	2023-01-01	+5.9%	47	53.2%	1.28	+2.103	-19.55%
W4	2023-01-01	2023-03-01	+11.5%	70	60.0%	1.42	+2.647	-21.01%
W5	2023-03-01	2023-05-01	+18.4%	50	64.0%	1.82	+4.064	-32.34%
W6	2023-05-01	2023-07-01	-1.6%	80	50.0%	0.97	+3.974	-28.36%
W7	2023-07-01	2023-09-01	-6.0%	64	50.0%	0.77	+2.623	-22.99%
W8	2023-09-01	2023-11-01	+7.9%	56	55.4%	1.28	+1.362	-13.11%
W9	2023-11-01	2024-01-01	+0.7%	69	50.7%	1.01	+0.416	-8.28%
W10	2024-01-01	2024-03-01	+24.8%	70	65.7%	1.90	+2.169	-23.58%
W11	2024-03-01	2024-05-01	+1.9%	63	55.6%	1.02	+4.915	-33.24%
W12	2024-05-01	2024-07-01	-22.0%	68	44.1%	0.74	+2.451	-22.51%
W13	2024-07-01	2024-09-01	+11.4%	78	57.7%	1.31	+3.187	-22.28%

Win	Train End	OOS End	OOS Ret	Trades	WR	PF	Sharpe	Max DD
W14	2024-09-01	2024-11-01	+8.8%	71	57.7%	1.26	+1.416	-13.32%
W15	2024-11-01	2025-01-01	+4.0%	69	53.6%	1.15	+2.225	-19.85%
W16	2025-01-01	2025-03-01	+1.0%	72	51.4%	1.00	+3.576	-26.04%
W17	2025-03-01	2025-05-01	+0.2%	78	53.8%	1.03	+2.088	-15.17%
W18	2025-05-01	2025-07-01	+10.3%	73	60.3%	1.33	+1.895	-15.82%
W19	2025-07-01	2025-09-01	-4.2%	73	49.3%	0.89	+1.784	-16.44%
W20	2025-09-01	2025-11-01	-3.9%	79	53.2%	0.92	+0.642	-9.26%
W21	2025-11-01	2026-01-01	+11.6%	72	56.9%	1.29	+1.723	-13.91%
W22	2026-01-01	2026-03-01	+21.2%	78	61.5%	1.61	+2.788	-22.22%
W23	2026-03-01	2026-05-01	-2.0%	73	53.4%	0.96	+3.833	-26.37%

Monte Carlo — In-Sample

1,000

Simulations

99.8%

P(Profit)

+80.0%

p5 Return

+274.0%

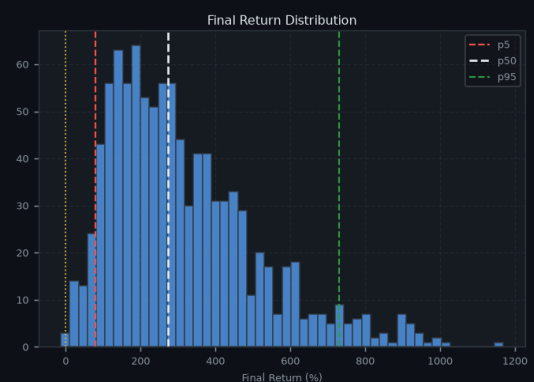
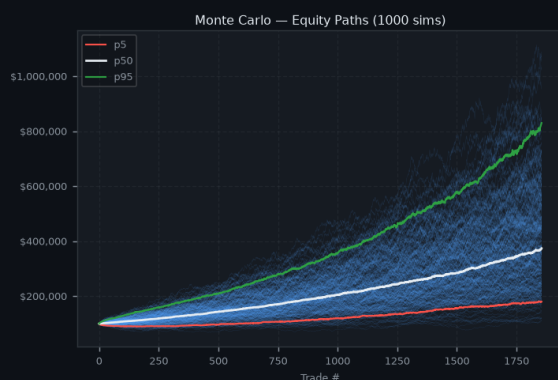
p50 Return

+730.1%

p95 Return

0.0%

P(Ruin >50%DD)



Monte Carlo — OOS Trades

1,000

Simulations

99.1%

P(Profit)

+26.9%

p5 Return

+123.1%

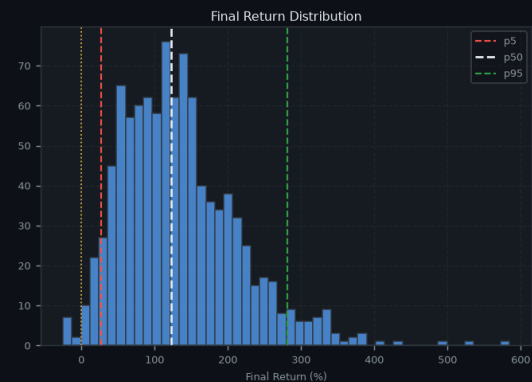
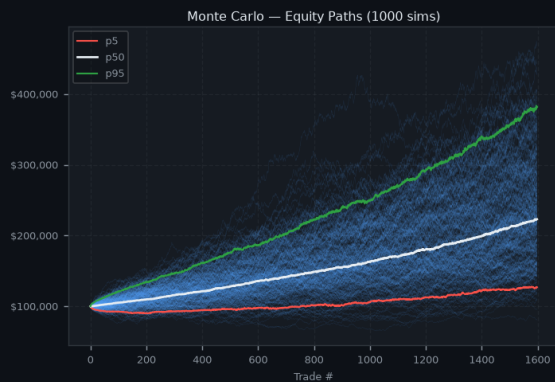
p50 Return

+281.9%

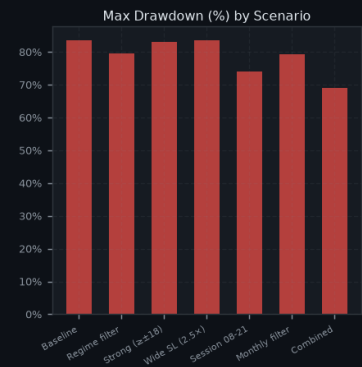
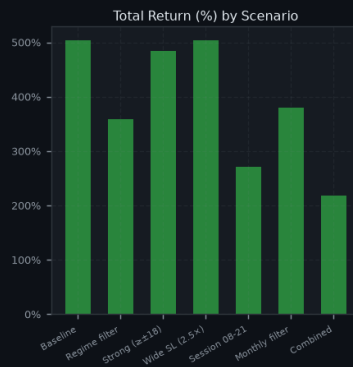
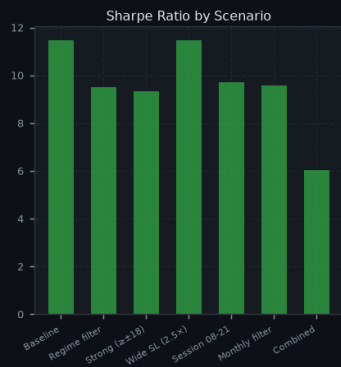
p95 Return

0.0%

P(Ruin >50%DD)



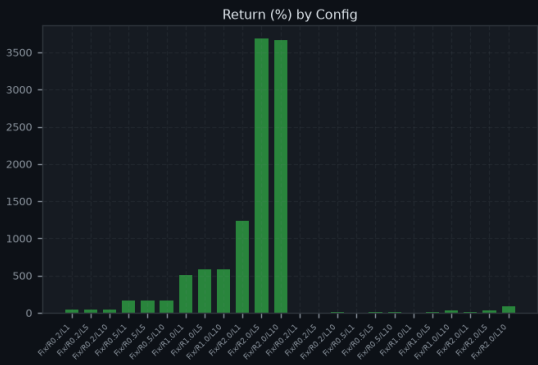
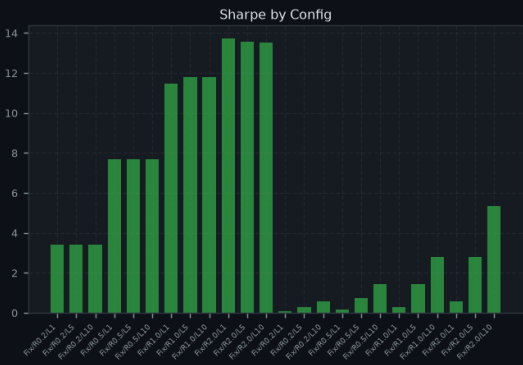
Scenario Comparison



Scenario	Return	Sharpe	Sortino	Calmar	Max DD	WR	PF	Trades
Baseline	+504.2%	11.476	34.242	6.029	-83.6%	56.7%	1.21	1,916
Regime filter	+359.5%	9.518	21.831	4.521	-79.5%	56.2%	1.20	1,658
Strong ($\geq \pm 18$)	+483.6%	9.340	22.373	5.824	-83.1%	58.4%	1.31	1,430
Wide SL (2.5x)	+504.2%	11.476	34.242	6.029	-83.6%	56.7%	1.21	1,916

Scenario	Return	Sharpe	Sortino	Calmar	Max DD	WR	PF	Trades
Session 08-21	+271.3%	9.720	19.589	3.665	-74.0%	55.5%	1.15	1,855
Monthly filter	+380.3%	9.566	21.795	4.790	-79.4%	57.3%	1.23	1,509
Combined	+218.7%	6.017	7.776	3.163	-69.2%	58.9%	1.31	844

Leverage × Sizing Grid



Scenario	Method	Pct (%)	Leverage	Total Ret (%)	Sharpe	Max DD (%)	Win Rate (%)
Baseline	Fixed Risk	0.2	1	+49.8%	3.433	-33.4%	56.2%
Baseline	Fixed Risk	0.2	5	+49.8%	3.433	-33.4%	56.2%
Baseline	Fixed Risk	0.2	10	+49.8%	3.433	-33.4%	56.2%
Baseline	Fixed Risk	0.5	1	+169.8%	7.689	-63.1%	56.2%
Baseline	Fixed Risk	0.5	5	+169.8%	7.692	-63.1%	56.2%
Baseline	Fixed Risk	0.5	10	+169.8%	7.692	-63.1%	56.2%
Baseline	Fixed Risk	1.0	1	+504.2%	11.476	-83.6%	56.2%
Baseline	Fixed Risk	1.0	5	+587.8%	11.801	-85.6%	56.2%
Baseline	Fixed Risk	1.0	10	+587.8%	11.801	-85.6%	56.2%

	Scenario	Method	Pct (%)	Leverage	Total Ret (%)	Sharpe	Max DD (%)	Win Rate (%)
Baseline	Baseline	Fixed Risk	2.0	1	+1233.0%	13.717	-92.8%	56.3%
	Baseline	Fixed Risk	2.0	5	+3682.6%	13.551	-97.4%	56.3%
	Baseline	Fixed Risk	2.0	10	+3667.8%	13.547	-97.4%	56.3%
	Baseline	Fixed Fraction	0.2	1	+0.6%	0.07	-0.6%	56.3%
	Baseline	Fixed Fraction	0.2	5	+3.1%	0.299	-3.1%	56.3%
	Baseline	Fixed Fraction	0.2	10	+6.4%	0.584	-6.1%	56.3%
	Baseline	Fixed Fraction	0.5	1	+1.6%	0.156	-1.6%	56.3%
	Baseline	Fixed Fraction	0.5	5	+8.1%	0.727	-7.5%	56.3%
	Baseline	Fixed Fraction	0.5	10	+16.7%	1.433	-14.4%	56.3%
	Baseline	Fixed Fraction	1.0	1	+3.1%	0.299	-3.1%	56.3%
	Baseline	Fixed Fraction	1.0	5	+16.7%	1.433	-14.4%	56.3%
	Baseline	Fixed Fraction	1.0	10	+35.9%	2.812	-26.6%	56.3%
	Baseline	Fixed Fraction	2.0	1	+6.4%	0.584	-6.1%	56.3%
	Baseline	Fixed Fraction	2.0	5	+35.9%	2.812	-26.6%	56.3%
	Baseline	Fixed Fraction	2.0	10	+83.4%	5.364	-45.8%	56.3%
	Session 08-21	Fixed Risk	0.2	1	+35.6%	2.661	-26.8%	55.9%
	Session 08-21	Fixed Risk	0.2	5	+35.6%	2.661	-26.8%	55.9%
	Session 08-21	Fixed Risk	0.2	10	+35.6%	2.661	-26.8%	55.9%

	Scenario	Method	Pct (%)	Leverage	Total Ret (%)	Sharpe	Max DD (%)	Win Rate (%)
	Session 08-21	Fixed Risk	0.5	1	+110.4%	6.143	-53.3%	55.9%
	Session 08-21	Fixed Risk	0.5	5	+110.7%	6.155	-53.4%	55.9%
	Session 08-21	Fixed Risk	0.5	10	+110.7%	6.155	-53.4%	55.9%
	Session 08-21	Fixed Risk	1.0	1	+271.3%	9.72	-74.0%	55.9%
	Session 08-21	Fixed Risk	1.0	5	+320.1%	10.204	-77.0%	55.9%
	Session 08-21	Fixed Risk	1.0	10	+320.1%	10.204	-77.0%	55.9%
	Session 08-21	Fixed Risk	2.0	1	+510.6%	12.577	-84.9%	55.9%
	Session 08-21	Fixed Risk	2.0	5	+1315.4%	13.312	-93.5%	55.9%
	Session 08-21	Fixed Risk	2.0	10	+1315.4%	13.312	-93.5%	55.9%
	Session 08-21	Fixed Fraction	0.2	1	+0.5%	0.058	-0.5%	55.9%
	Session 08-21	Fixed Fraction	0.2	5	+2.3%	0.247	-2.4%	55.9%
	Session 08-21	Fixed Fraction	0.2	10	+4.7%	0.483	-4.7%	55.9%
	Session 08-21	Fixed Fraction	0.5	1	+1.2%	0.129	-1.2%	55.9%
	Session 08-21	Fixed Fraction	0.5	5	+6.0%	0.601	-5.8%	55.9%
	Session 08-21	Fixed Fraction	0.5	10	+12.2%	1.185	-11.3%	55.9%
	Session 08-21	Fixed Fraction	1.0	1	+2.3%	0.247	-2.4%	55.9%
	Session 08-21	Fixed Fraction	1.0	5	+12.2%	1.185	-11.3%	55.9%
	Session 08-21	Fixed Fraction	1.0	10	+25.7%	2.331	-21.2%	55.9%

	Scenario	Method	Pct (%)	Leverage	Total Ret (%)	Sharpe	Max DD (%)	Win Rate (%)
	Session 08-21	Fixed Fraction	2.0	1	+4.7%	0.483	-4.7%	55.3%
	Session 08-21	Fixed Fraction	2.0	5	+25.7%	2.331	-21.2%	55.3%
	Session 08-21	Fixed Fraction	2.0	10	+57.0%	4.499	-37.6%	55.3%
	Regime filter	Fixed Risk	0.2	1	+39.1%	2.526	-28.9%	56.3%
	Regime filter	Fixed Risk	0.2	5	+39.1%	2.526	-28.9%	56.3%
	Regime filter	Fixed Risk	0.2	10	+39.1%	2.526	-28.9%	56.3%
	Regime filter	Fixed Risk	0.5	1	+125.5%	5.835	-56.9%	56.3%
	Regime filter	Fixed Risk	0.5	5	+124.9%	5.799	-56.8%	56.3%
	Regime filter	Fixed Risk	0.5	10	+124.9%	5.799	-56.8%	56.3%
	Regime filter	Fixed Risk	1.0	1	+359.5%	9.518	-79.5%	56.3%
	Regime filter	Fixed Risk	1.0	5	+381.3%	9.407	-80.4%	56.3%
	Regime filter	Fixed Risk	1.0	10	+381.3%	9.407	-80.4%	56.3%
	Regime filter	Fixed Risk	2.0	1	+935.8%	12.193	-91.5%	56.3%
	Regime filter	Fixed Risk	2.0	5	+1818.1%	11.69	-95.4%	56.3%
	Regime filter	Fixed Risk	2.0	10	+1800.4%	11.669	-95.3%	56.3%
	Regime filter	Fixed Fraction	0.2	1	+0.6%	0.062	-0.6%	56.3%
	Regime filter	Fixed Fraction	0.2	5	+2.9%	0.259	-2.9%	56.3%
	Regime filter	Fixed Fraction	0.2	10	+5.9%	0.505	-5.8%	56.3%

Scenario	Method	Pct (%)	Leverage	Total Ret (%)	Sharpe	Max DD (%)	Win Rate (%)
Regime filter	Fixed Fraction	0.5	1	+1.4%	0.136	-1.5%	56.3%
Regime filter	Fixed Fraction	0.5	5	+7.4%	0.627	-7.2%	56.3%
Regime filter	Fixed Fraction	0.5	10	+15.3%	1.236	-13.8%	56.3%
Regime filter	Fixed Fraction	1.0	1	+2.9%	0.259	-2.9%	56.3%
Regime filter	Fixed Fraction	1.0	5	+15.3%	1.236	-13.8%	56.3%
Regime filter	Fixed Fraction	1.0	10	+32.6%	2.425	-25.6%	56.3%
Regime filter	Fixed Fraction	2.0	1	+5.9%	0.505	-5.8%	56.3%
Regime filter	Fixed Fraction	2.0	5	+32.6%	2.425	-25.6%	56.3%
Regime filter	Fixed Fraction	2.0	10	+74.8%	4.642	-44.3%	56.3%